

TRANSFORMATION: STRUCTURING

What is Structuring?

- Changes the **form, schema (column/row layout) or granularity (level of detail)** of existing data.
- Divided into **Intrarecord** and **Interrecord** structuring.

1. Intrarecord Structuring

Works with individual records(rows) and fields(columns).

A. Reordering Record Fields

- Changes the **order of columns/fields** within a record/rows.
- The data values remain identical; only their arrangement changes.

B. Extracting Values

- Creates a **new field from an existing field** by extracting useful information.
 - **Simple Positional Extraction** - extracts a substring using its fixed starting/ending position.
 - **Complex Positional Extraction** - extracts substrings of varying lengths separated by a consistent **delimiter** (symbol or character that acts as a separator between different pieces of data) character.
 - **Pattern-Based Extraction** - extracts a substring identified by a generic pattern.
 - **Complex Structure Extraction** - extracts individual elements from **JSON maps** (A collection of key-value pairs enclosed in curly brackets {}, where the key represents a property name and the value represents that property's content) **or JSON arrays** (An ordered sequence of values enclosed in square brackets [], with elements separated by commas and enclosed in double quotes)
Eg - ["Sally","Bob","Alon","Georgia"] and {"product":"Trifacta Wrangler","price":"free","category":"wrangling tool"}

C. Combining Multiple Record Fields

- Merges related fields into **one field**. It is the exact reverse of extraction.
- Example: Combining a `City` column (`MCPHERSON`) and a `State` column (`KS`) into a single column separated by a comma and space: `MCPHERSON, KS`

2. Interrecord Structuring

Works across multiple records and fields.

A. Filtering Records and Fields

Used for **data cleaning** and to **alter the granularity** and scope of a dataset

- **Record-Based Filtering** — removes selected **records/rows**.
- **Field-Based Filtering** — removes selected **fields/columns**.

B. Aggregations and Pivots

Used to **shift the granularity** and organization of the dataset.

Simple Aggregation

- Combines **multiple input records** into **summarized output records**.
- Common operations:
SUM, MEAN, MIN, COUNT, List Concatenation
- Example: contributions can be summarized **by campaign committee**.

Column-to-Row Pivot

- **One input record** can produce **multiple output records**.
- Values from columns become characteristics of new rows.
- Also called **Unpivoting**..

Row-to-Column Pivot

- Multiple input records can produce multiple output records.
- Converts values from records into **new fields/columns**.
- Also called **Pivoting**.